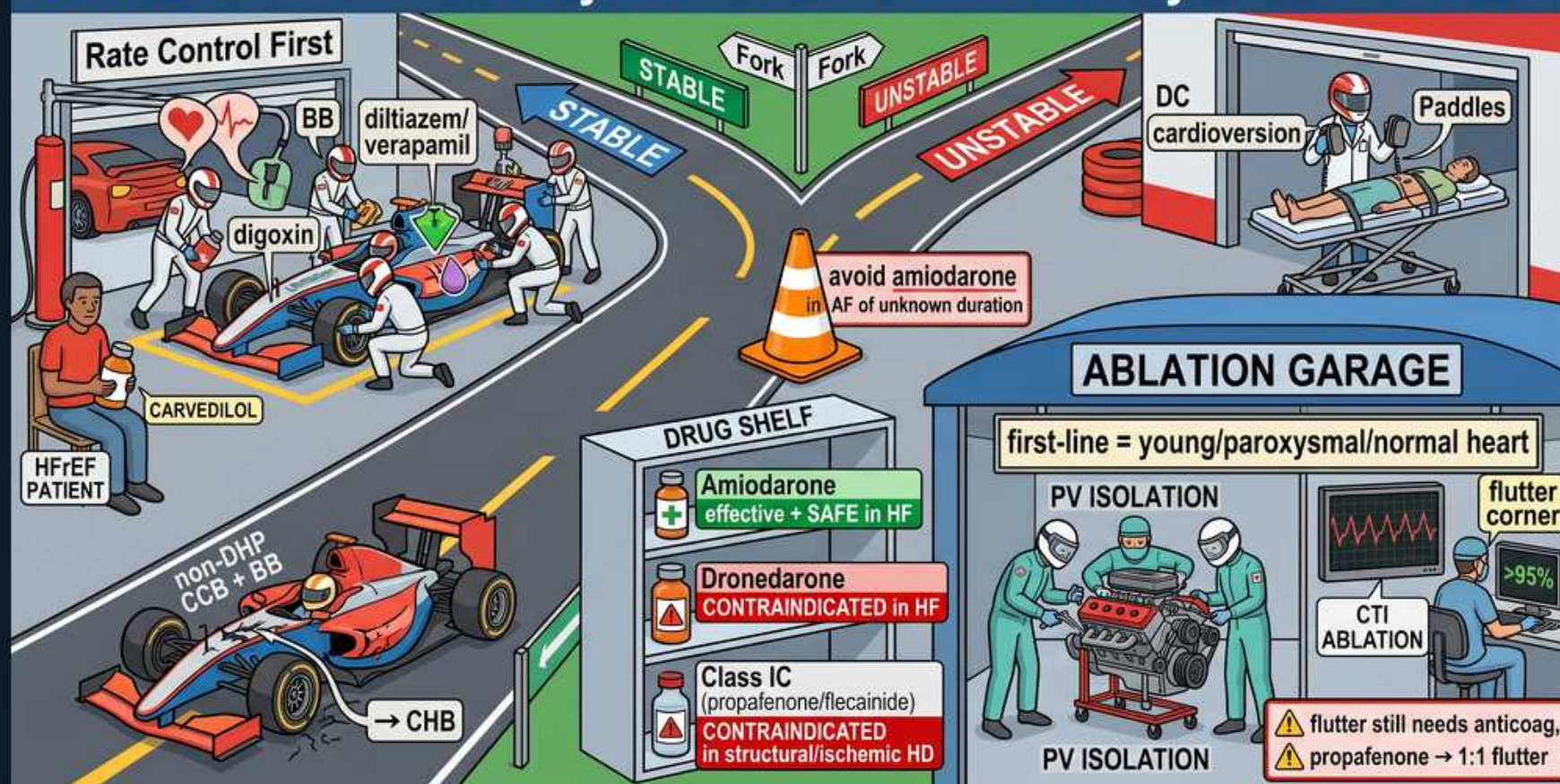


SCENE 2 — AF Rate/Rhythm & Ablation: “The Rhythm Racetrack”



1. Rate control first (stable)

WHAT YOU SEE

'Rate Control First' pit crew with BB, diltiazem/verapamil, digoxin.

WHAT IT ENCODES

In a hemodynamically stable patient, rate control is the initial strategy: β -blockers or non-dihydropyridine CCBs (diltiazem/verapamil) first-line, digoxin as add-on (especially with heart failure or hypotension). RACE-II showed lenient (<110) rate control is non-inferior to strict in many patients.

BOARD TRAP

WRONG: 'every new AF needs urgent rhythm control.' Stable patients are rate-controlled first; rhythm control is for persistent symptoms or specific indications.

2. Unstable → DC cardioversion

WHAT YOU SEE

'UNSTABLE' fork → DC cardioversion with paddles.

WHAT IT ENCODES

Hemodynamic instability (hypotension, ischemia, pulmonary edema, shock) from AF mandates immediate synchronized DC cardioversion regardless of duration — you don't wait for anticoagulation when the patient is crashing.

BOARD TRAP

WRONG: 'rate-control drugs first even if hypotensive.' AV-nodal blockers worsen shock; unstable AF needs electrical cardioversion now.

3. Avoid amiodarone if unknown duration

WHAT YOU SEE

Cone: 'avoid amiodarone in AF of unknown duration'.

WHAT IT ENCODES

Amiodarone can chemically cardiovert AF; giving it when AF duration is >48 h or unknown (without anticoagulation/TEE) risks dislodging an atrial thrombus → embolic stroke.

BOARD TRAP

WRONG: 'amiodarone is a safe rate-control choice in any AF.' Its cardioverting effect makes it unsafe in unanticoagulated AF of uncertain duration.

4. Amiodarone — effective + SAFE in HF

WHAT YOU SEE

Drug shelf: Amiodarone, green, 'effective + SAFE in HF'.

WHAT IT ENCODES

Amiodarone is the most effective agent for maintaining sinus rhythm and is the one antiarrhythmic considered SAFE in structural heart disease and HFrEF (along with dofetilide). Long-term use carries thyroid, pulmonary, hepatic, and skin toxicity.

BOARD TRAP

WRONG: 'amiodarone is contraindicated in heart failure.' It is the preferred rhythm-control drug in HF; class IC and dronedarone are the ones to avoid.

5. Dronedarone — CONTRAINDICATED in HF

WHAT YOU SEE

Shelf: Dronedarone, red, CONTRAINDICATED in HF.

WHAT IT ENCODES

Dronedarone is contraindicated in decompensated/NYHA III-IV heart failure and permanent AF — ANDROMEDA and PALLAS trials showed increased mortality. Use only in lower-risk paroxysmal/persistent AF without significant HF.

BOARD TRAP

WRONG: 'dronedarone is a safer amiodarone for HF patients.' It increases mortality in HF/permanent AF.

6. Class IC — CONTRAINDICATED in structural HD

WHAT YOU SEE

Shelf: Class IC (propafenone/flecainide), red, CONTRAINDICATED in structural/ischemic HD.

WHAT IT ENCODES

Class IC agents (flecainide, propafenone) are only for structurally normal hearts; in ischemic or structural heart disease they are proarrhythmic and increase mortality (CAST trial). Often combined with an AV-nodal blocker to prevent 1:1 flutter conduction.

BOARD TRAP

WRONG: 'flecainide is fine post-MI.' Class IC drugs are contraindicated in ischemic/structural heart disease.

7. Non-DHP CCB + BB → CHB

WHAT YOU SEE

Crashed car: non-DHP CCB + β -blocker → complete heart block.

WHAT IT ENCODES

Combining a non-dihydropyridine CCB (verapamil/diltiazem) with a β -blocker stacks AV-nodal suppression → severe bradycardia, high-grade AV block, and hypotension.

BOARD TRAP

WRONG: 'verapamil plus metoprolol gives smoother rate control.' The combination risks complete heart block — avoid.

8. HFrEF patient → carvedilol

WHAT YOU SEE

HFrEF patient with carvedilol.

WHAT IT ENCODES

In AF with HFrEF, a β -blocker (e.g., carvedilol) provides rate control and mortality benefit and helps reverse tachycardia-induced cardiomyopathy. Avoid non-DHP CCBs (negative inotropes) in reduced EF.

BOARD TRAP

WRONG: 'use diltiazem for rate control in HFrEF.' Non-DHP CCBs are negative inotropes — contraindicated in reduced EF; use a β -blocker.

9. Ablation first-line = young/paroxysmal/normal heart

WHAT YOU SEE

Ablation garage sign: first-line = young/paroxysmal/normal heart.

WHAT IT ENCODES

Catheter (pulmonary-vein isolation) ablation is most effective — and increasingly first-line — in younger, symptomatic patients with paroxysmal AF and a structurally normal heart. In HFrEF, PV isolation improves outcomes (CASTLE-AF).

BOARD TRAP

WRONG: 'ablation works best for long-standing persistent AF in a dilated atrium.' Success is highest in paroxysmal AF with a normal-sized atrium.

10. PV isolation

WHAT YOU SEE

Surgeons performing PV isolation on a heart.

WHAT IT ENCODES

Pulmonary-vein isolation electrically disconnects the PV triggers that initiate AF. Complications to know: PV stenosis, atrio-esophageal fistula, cardiac tamponade, phrenic nerve injury.

BOARD TRAP

WRONG: 'ablation cures AF permanently in one procedure for everyone.' Recurrence and repeat procedures are common, especially in persistent AF.

11. Typical flutter → CTI ablation >95%

WHAT YOU SEE

Flutter corner: sawtooth waves, CTI ablation, >95%.

WHAT IT ENCODES

Typical (cavotricuspid-isthmus-dependent) atrial flutter is cured by CTI ablation with >95% success — definitive therapy preferred over long-term antiarrhythmics. Flutter still carries AF-equivalent stroke risk and needs anticoagulation by CHA₂DS₂-VASc.

BOARD TRAP

WRONG: 'atrial flutter doesn't need anticoagulation.' Flutter carries the same thromboembolic risk as AF.

12. Propafenone → 1:1 flutter risk

WHAT YOU SEE

Warning: propafenone → 1:1 flutter conduction.

WHAT IT ENCODES

Class IC drugs can slow the atrial flutter rate enough that the AV node conducts 1:1 → dangerously fast ventricular rates. Co-administer an AV-nodal blocker (β-blocker/CCB) when using class IC for AF/flutter.

BOARD TRAP

WRONG: 'slowing the atrial rate with flecainide always slows the ventricles.' Slower flutter can conduct 1:1, paradoxically speeding the ventricular rate.
